

AQI Analysis & Prediction

Pollution Trends and Health Impact in Delhi, Mumbai, Kolkata, Bengaluru and Chennai (2023–2024)

This report examines two years of daily air quality readings from five Indian cities to answer questions: which city has the worst air, how strongly pollution is with respiratory illness, which pollutants matter most, and whether AQI can be predicted from just raw pollutant readings . The dataset covers **3,655 daily records** across 2023–2024, with seven pollutants measured : PM2.5, PM10, NO2, NH3, SO2, CO, and O3, with a severity label running from Good to Severe.

Dataset Info (Official government data)

Field	Detail
Cities Covered	Delhi, Mumbai, Kolkata, Bengaluru, Chennai
Time Span	January 2023 – December 2024
Daily Records	3,655
Pollutants	PM2.5, PM10, NO2, NH3, SO2, CO, O3
Severity Bands	Good, Satisfactory, Moderate, Poor, Very Poor, Severe

3,655

Daily Records

Across both years

7

Pollutants Measured

5

Cities Analyzed

Major cities

6

Severity Bands

Good to Severe

Which City Has the Worst Air?

Averaged for both years, Delhi sits above the rest at a mean AQI of **217.9**, placing it in the "Poor" band – more than 2.5 times Chennai's average of 83.7, the cleanest city in this dataset. Kolkata and Mumbai fall in between, both in "Moderate" territory, while Bengaluru and Chennai stay closest to the "Satisfactory" range. The gap between Delhi and every other city is the single largest signal in the entire dataset.

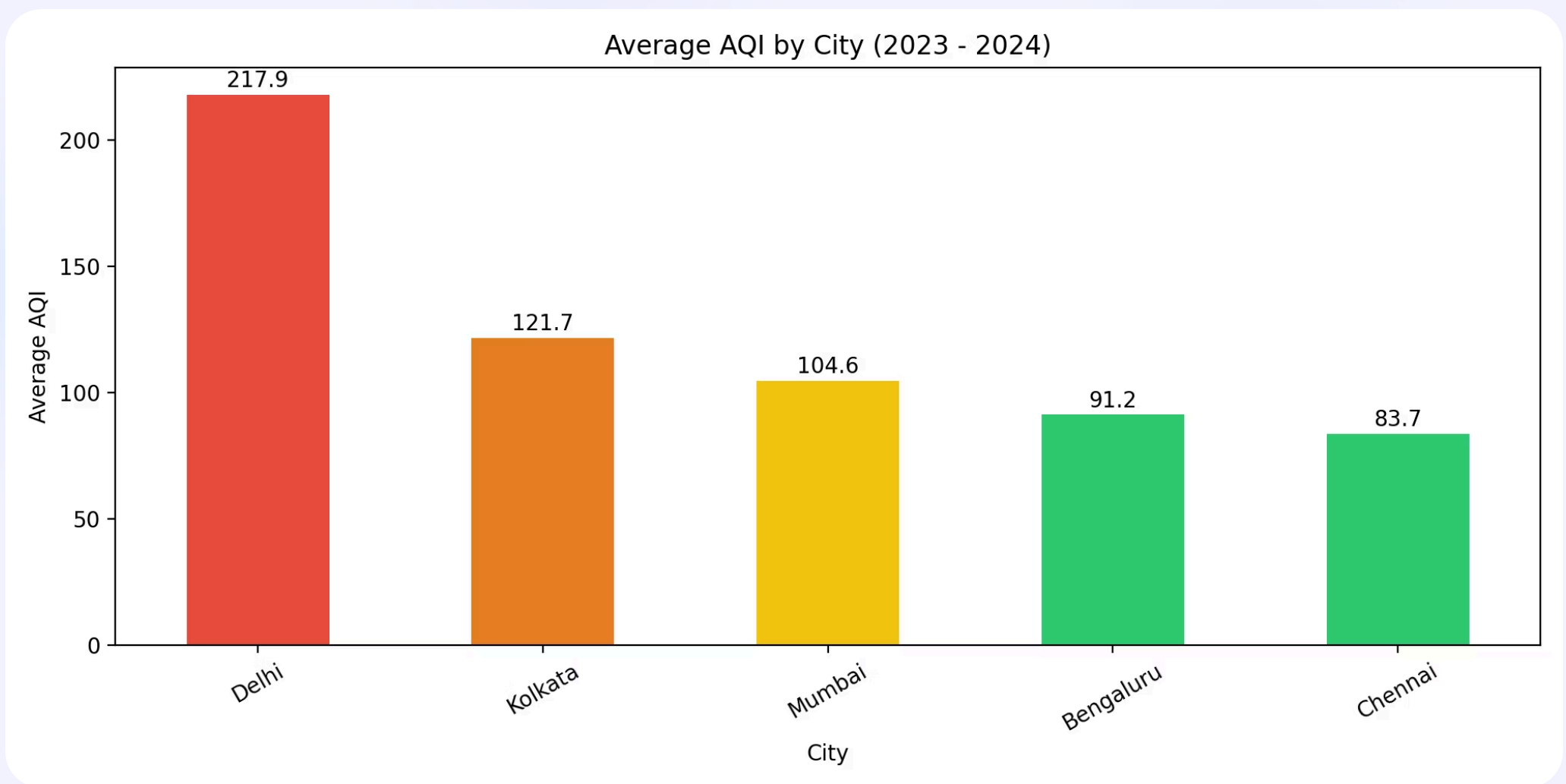


Figure 1 – Average AQI by city, 2023–2024

Does Worse Air Actually Mean More Sickness?

The next step was to check whether respiratory case counts climb with AQI severity. The boxplot below groups daily respiratory case counts by the severity label assigned to that day.

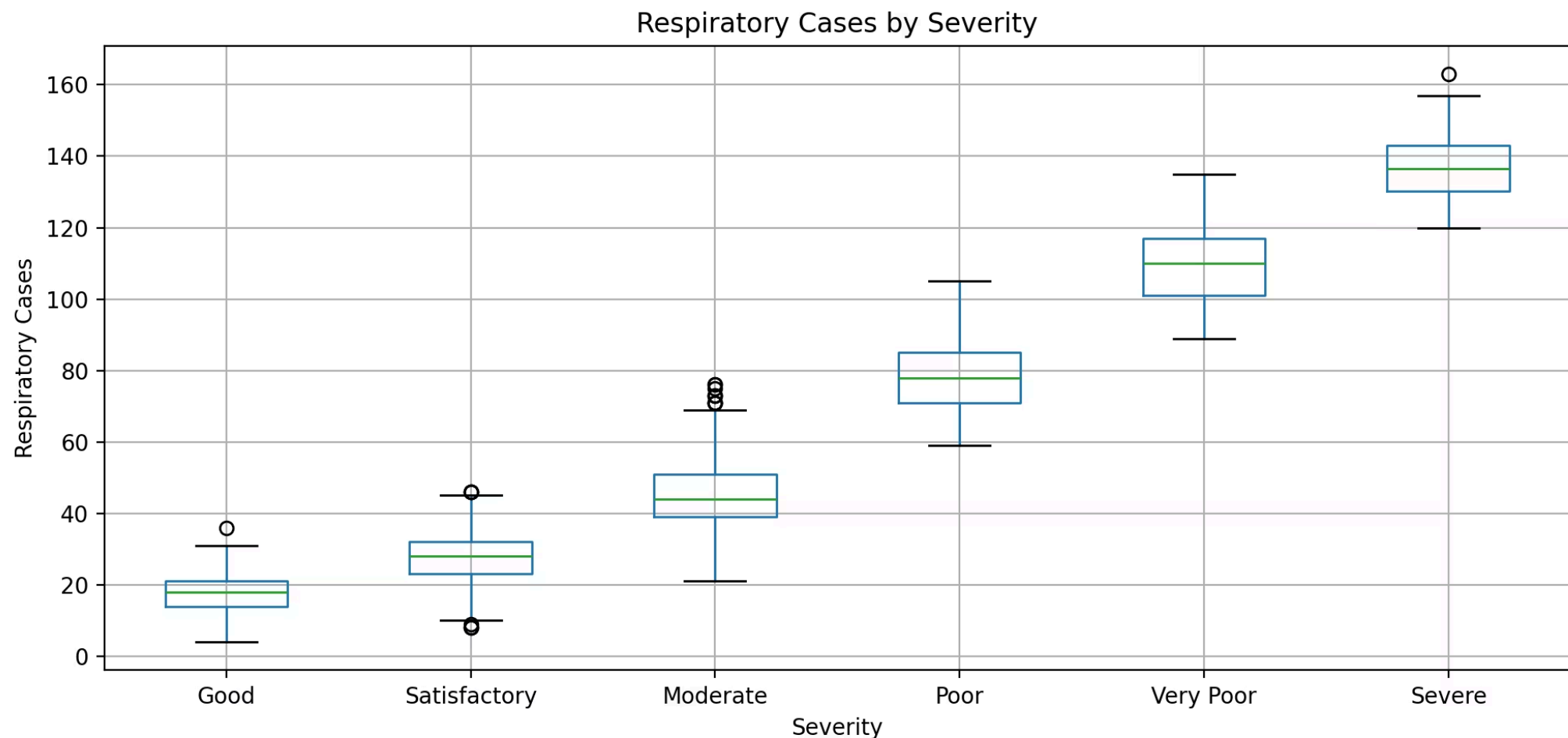


Figure 2 – Distribution of respiratory cases across severity bands

Average respiratory cases climb steadily from **17.2 on Good days** to **136.8 on Severe days** – roughly an eightfold increase. There's no ambiguity here: as the air gets worse, hospitals see it.

Which Pollutant Actually Drives AQI?

AQI is a composite score built from seven separate pollutant readings, but they don't all contribute equally. Running a correlation between each pollutant and the overall AQI score shows which ones are doing the heavy lifting.

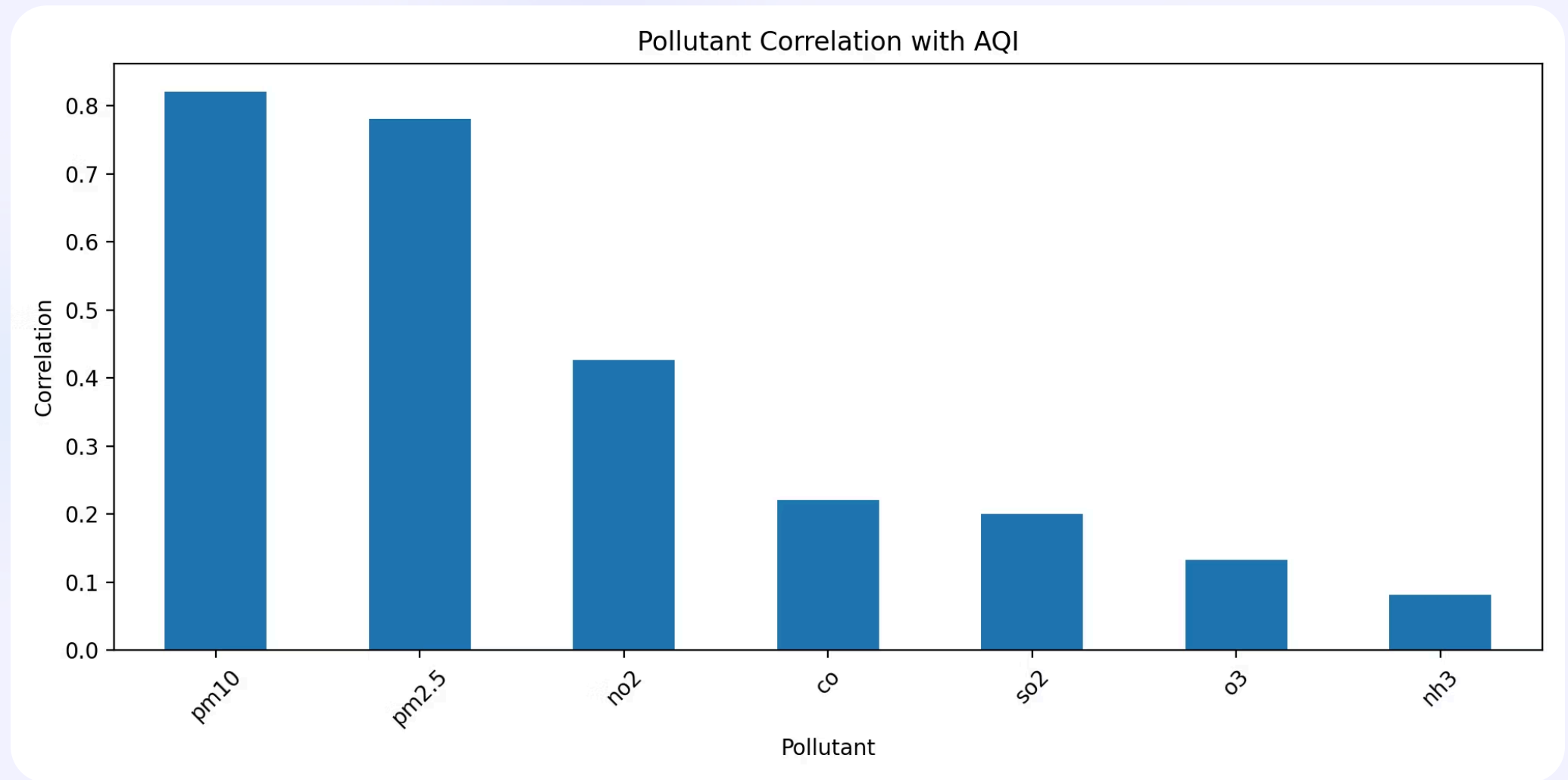


Figure 3 – Correlation of each pollutant with AQI

Particulate matter dominates the picture. **PM10** correlates with AQI at **0.82** and **PM2.5** at **0.78**, both far ahead of NO2 (0.43) and the remaining gaseous pollutants, which all sit below 0.25.

Is Pollution Worse in Certain Months?

Pollution isn't constant throughout the year – wind patterns, temperature, and crop burning all shift seasonally. Averaging AQI by month across both years brings out the seasonal shape clearly.

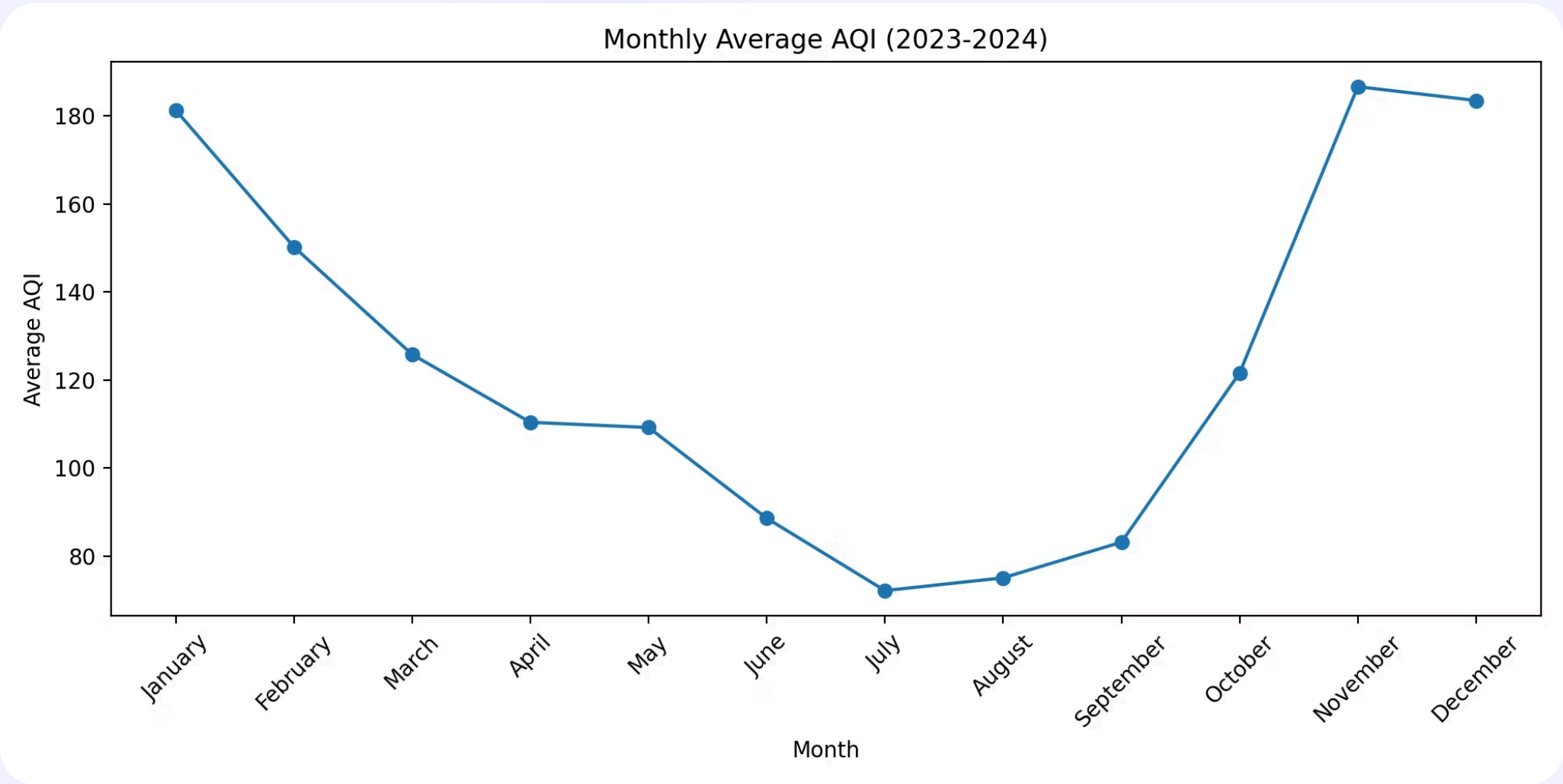


Figure 4 – Average AQI by month, pooled across 2023 and 2024

Winter Peak

AQI climbs through autumn and peaks in **November at 186.6**, staying elevated through December (183.5) and January (181.3). Calmer winds and temperature inversion trap pollutants closer to the ground.

Summer Low

AQI falls steadily through spring and summer, bottoming out in **July at 72.1** before starting to climb again in autumn. Monsoon rains help wash the air clean mid-year.

Is Delhi's Pollution Constant or Occasional?

Figure 1 showed Delhi's average is high, but an average alone can't tell you whether that's driven by a handful of extreme days or by consistently bad air. Counting how many days each city spent in each severity band answers that.

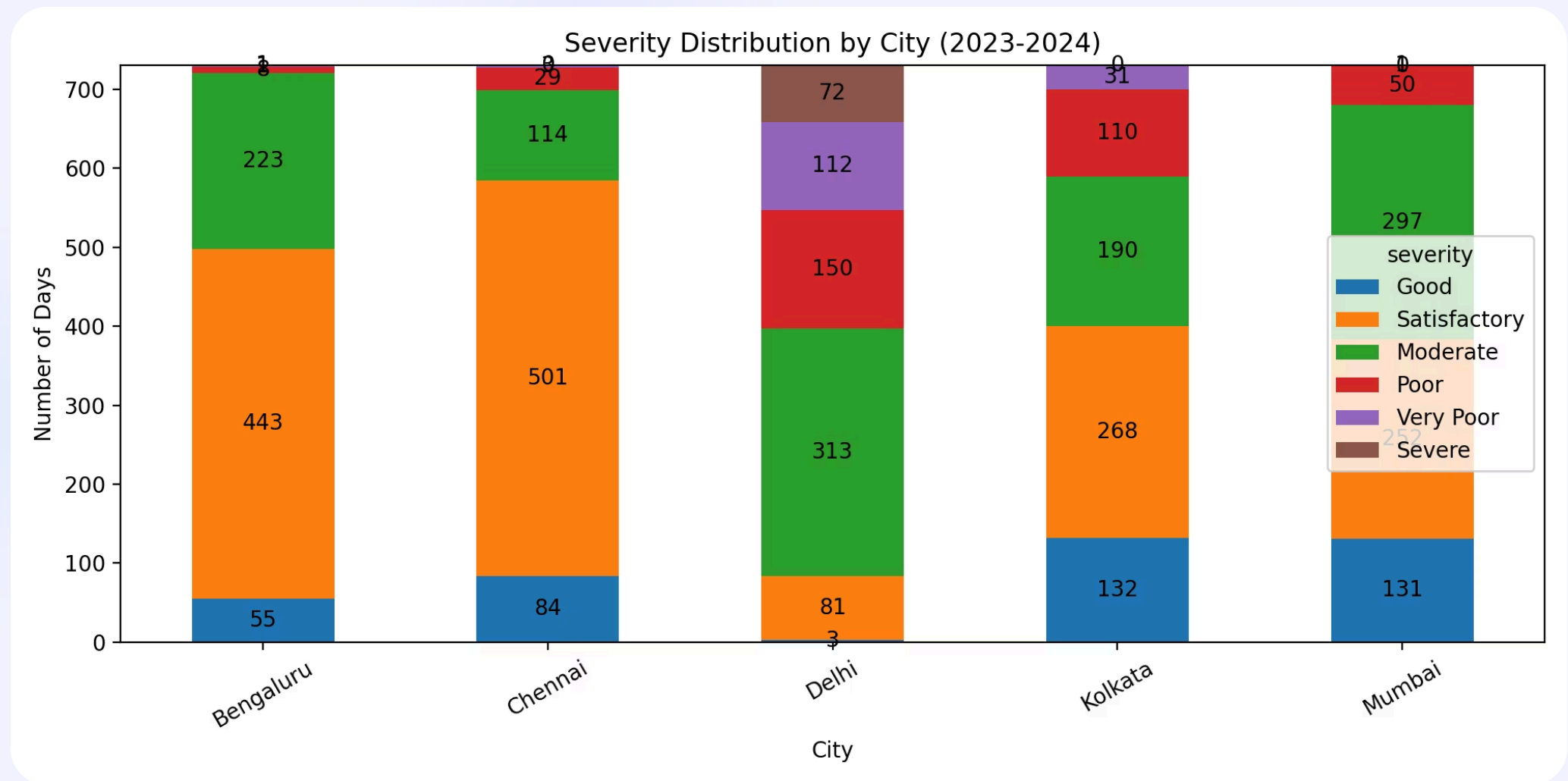


Figure 5 – Number of days spent in each severity band, by city

Delhi spends a large chunk of the year in **Poor territory or worse** – this isn't a few bad weeks dragging the average up, it's sustained exposure. Chennai and Bengaluru, by contrast, spend most of their days in the Satisfactory or Moderate bands and rarely cross into anything worse. This confirms that Delhi's elevated average reflects a genuinely different baseline air quality.

Is Air Quality Getting Better Over Time?

With two full years of data, it's possible to check whether any of these cities are actually improving, rather than just looking at a static snapshot. The chart below compares each city's average AQI in 2023 against 2024.

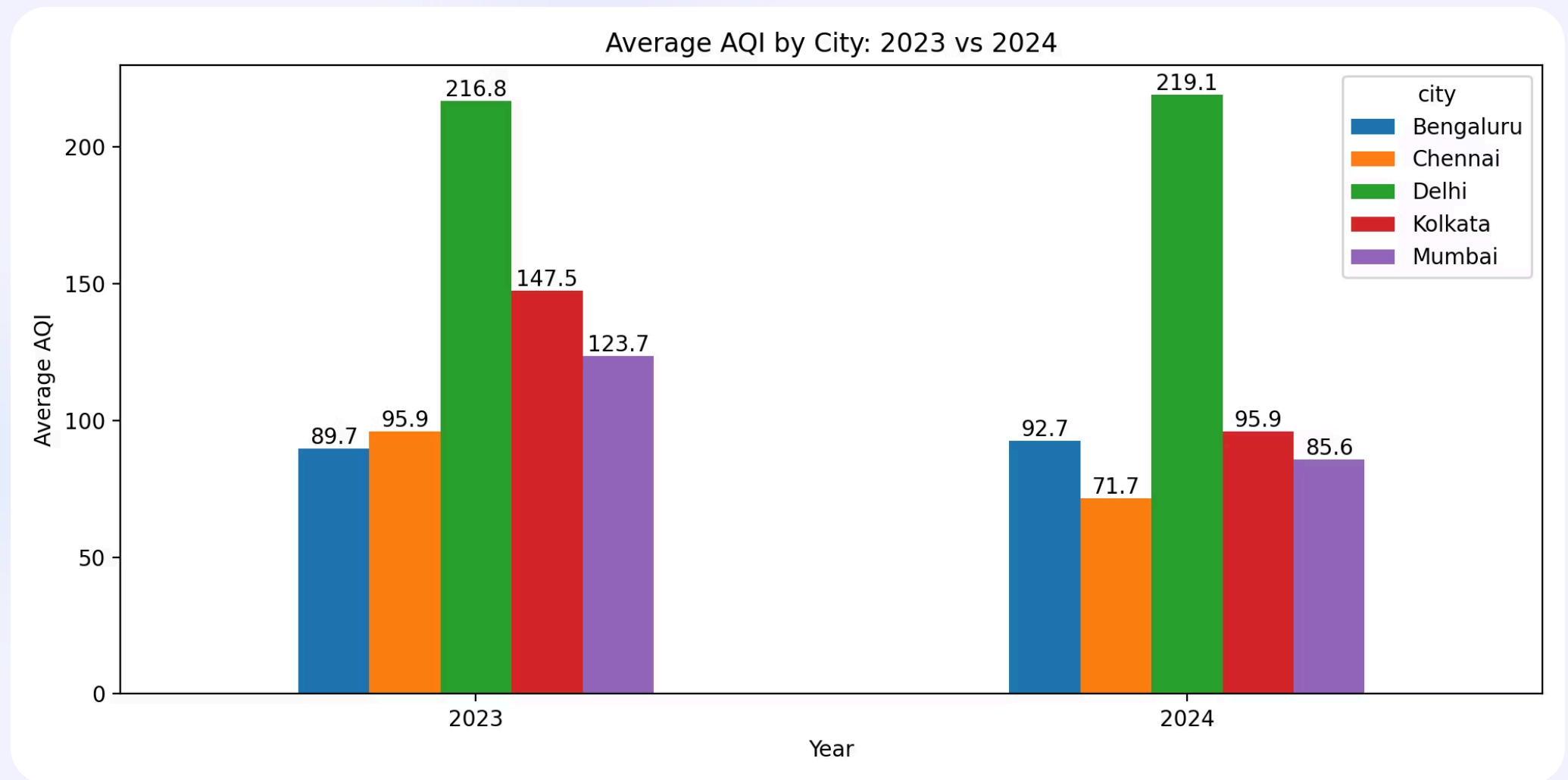


Figure 6 – Average AQI per city, comparing 2023 to 2024

Kolkata improved the most, dropping from 147.5 to 95.9 (a **35% reduction**), with **Mumbai** close behind at a **31% drop** (123.7 to 85.6) and **Chennai** improving by **25%** (95.9 to 71.7). Delhi and Bengaluru, however, barely moved – Delhi actually ticked up slightly (216.8 to 219.1) and Bengaluru rose marginally too (89.7 to 92.7). So three cities are making real progress, but the most polluted city in the dataset hasn't budged, which means the national picture can't be summarized as simply "improving" or "static."

Can AQI Be Predicted From Pollutant Levels Alone?

The final question is a practical one: if you only had the seven raw pollutant readings for a given day, how accurately could you reconstruct the AQI score? A Random Forest Regressor was trained on 80% of the data and tested on the remaining 20% it never saw during training.

Metric	Result	What it means
R ² Score	0.84	84% of the variation in AQI is explained by the seven pollutant inputs
Mean Absolute Error	23.2 points	Typical prediction is off by about 23 points, against AQI's 0–500 scale

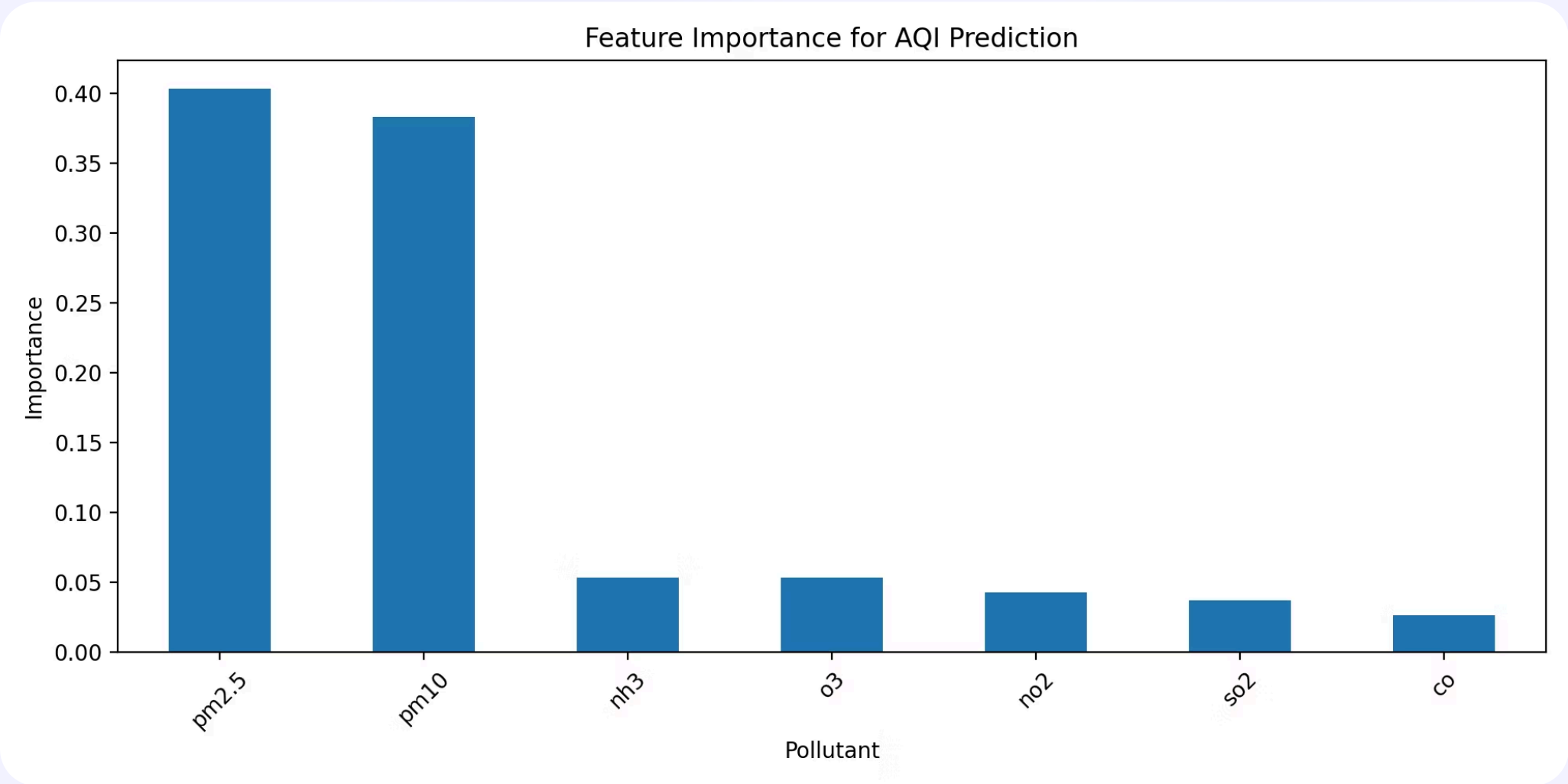


Figure 7 – Relative importance of each pollutant in the prediction model

PM2.5 (importance 0.40) and PM10 (0.38) again come out on top, consistent with the correlation findings in Figure 3 – though notably PM2.5 edges ahead of PM10 here, which it didn't in the simple correlation. That's a small but real difference: the model is accounting for overlap between the two correlated particulate measures, where a plain correlation treats each pollutant in isolation and can't separate their shared signal.

Random Forest was the right tool for this rather than a linear model for three reasons: the pollutant-to-AQI relationship isn't strictly linear, so a method that doesn't assume linearity has an advantage; it produces feature importance scores as a direct byproduct, which is exactly what's needed to answer "which pollutant matters most"; and it tends to be less thrown off by outliers, which matters here given how skewed the respiratory case counts were in Figure 2.

Summary of Findings

Delhi Is the Clear Outlier

Averaging an AQI of 217.9 – more than double every other city in the dataset – Delhi sits firmly in the "Poor" band while all others remain in Moderate or Satisfactory territory.

Near-Linear Health Impact

Respiratory cases rise close to eightfold between Good and Severe days, with a correlation of 0.98. As the air gets worse, hospitals see it.

Particulates Drive AQI

PM2.5 and PM10 dominate AQI far more than any gaseous pollutant. NO₂, SO₂, CO, and O₃ all play a distant second.

Delhi's Problem Is Chronic

The city spends a substantial share of the year in Poor or worse conditions – sustained exposure, not a few isolated bad days.

Progress Is Real but Uneven

Kolkata, Mumbai, and Chennai improved meaningfully (–35%, –31%, –25%), while Delhi and Bengaluru stayed essentially flat.

AQI Predictable From Pollutants

A Random Forest model reconstructs AQI with an R^2 of 0.84 and an average error of 23.2 points – strong, though not perfect.

📄 **Tools used:** Python, pandas, matplotlib, seaborn, scikit-learn (Random Forest Regression)